National Institute of Technology Jamshedpur
Department of Computer Science and Engineering

Research-Oriented B. Tech Capstone Proposal

A Case Study on NIT Jamshedpur's Academic & Administrative Query Assistant, Validated Across Institutions

Supervisor: Dr. Dilip Kumar
Semester: 5th Semester, B.Tech CSE
Team Size: 6

Student Name	Roll Number	Primary Responsibility
Sujal Kumar Yadav	2024UGCS033	Verification Agent Architecture & Orchestration
Tanishk S Pandey	2024UGCS037	LLM Response Generator & Retrieval (RAG) Layer
Mohit Anand	2024UGCS072	Groundedness Scoring Algorithm & Evaluation
Ankit Sharma	2024UGCS063	Fact-Check & Policy-Compliance Agents, Backend API
Dhirendra Pratap Dindoure	2024UGCS084	Web Dashboard & Results Visualization
Himanshu Gupta	2024UGCS113	Dataset Curation (NITJ + Public), Baselines & Human Evaluation

1. Abstract

Large Language Models (LLMs) are increasingly deployed as academic assistants in universities worldwide, but their tendency to hallucinate factual claims — incorrect exam dates, fabricated policy details, invented deadlines — poses real risk to students who trust these systems. Existing mitigation approaches (retrieval-augmented generation, self-consistency checking) reduce but do not eliminate ungrounded claims, and prior evaluations are typically confined to a single institution, weakening generalizability claims. This project proposes a domain-agnostic, multi-agent groundedness verification pipeline that extracts atomic claims from LLM-generated academic responses and verifies each against official institutional records, a deterministic policy rule base, and a retrieval-grounded document store. A novel groundedness scoring algorithm aggregates these signals into a per-claim confidence score driving a decision gate that passes, flags, or regenerates ungrounded content. The framework is validated on two independent datasets — a primary dataset built from NIT Jamshedpur's academic and administrative records, and a secondary public university dataset — against plain-LLM, RAG-only, and self-consistency baselines, with results reported via standard classification metrics (ROC/PR curves, confusion matrices) and human evaluation agreement. Cross-institution consistency in results is used to support the framework's generalizability beyond a single university deployment.

2. Problem Statement & Motivation

University academic assistants built on LLMs routinely blend genuinely retrieved facts with plausible-sounding but fabricated deadlines, policy details, or procedural claims. This is a global problem: any institution deploying an LLM-based assistant faces the same risk, yet most published evaluations validate against a single institution's data, leaving generalizability an open question. This project addresses both the technical gap (no widely-cited pipeline performs explicit, auditable, per-claim verification for institutional/academic claims) and the evaluation gap (cross-institution validation) in one design.

3. Novelty Statement

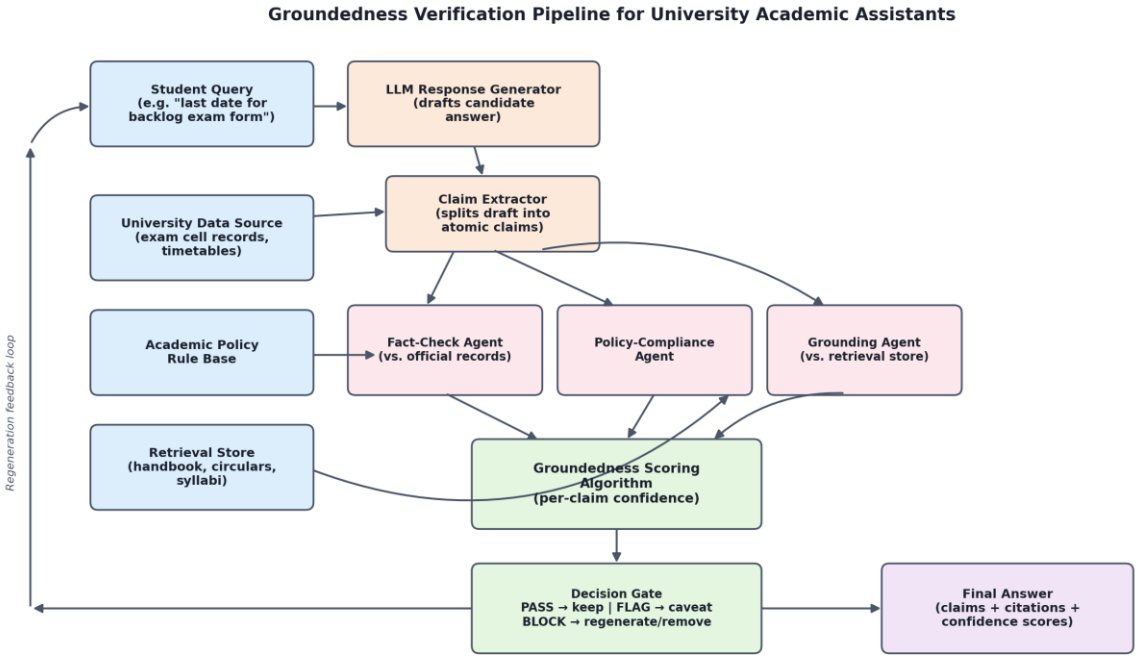
The core contribution is not the use of an LLM or a multi-agent framework in isolation — both are well established. The novelty is:

- A claim-level (not response-level) verification architecture decomposing generated text into atomic, independently verifiable claims before checking each one against institutional records, policy rules, and retrieval-grounded documents.
- A groundedness scoring algorithm aggregating three heterogeneous verification signals into a single auditable confidence score per claim — the aggregation method and its calibration is the primary technical contribution.
- A decision-gate mechanism with a regeneration feedback loop, closing the loop from detection back to correction.
- Cross-institution validation on two independent university datasets (NITJ + a public dataset) — directly answering the generalizability concern that limits most single-institution chatbot evaluation papers, and the key differentiator for SCI/Scopus reviewers.

4. Related Work & Gap Analysis

Existing Approach	What It Does	Gap This Project Addresses
RAG-based educational/institutional chatbots	Ground LLM answers in retrieved handbook/FAQ documents	No explicit per-claim verification against structured records (exam dates, fee deadlines) or policy rules
LLM hallucination detection (self-consistency, NLI-based checkers)	Detects hallucination via internal consistency or entailment models	General-purpose; not adapted to institutional-policy claims verifiable against authoritative records
Multi-agent LLM frameworks (planner-executor-critic)	Improves multi-step reasoning reliability	Critic step is usually another LLM judging plausibility, not a deterministic, auditable rule/data check
Single-institution chatbot evaluation studies	Report accuracy/satisfaction for one university's deployed assistant	Rarely validated across more than one institution, weakening generalizability claims
Responsible-AI / constitutional AI approaches	Constrain LLM behavior via principles or self-critique	Constraints are typically qualitative/behavioral, not quantitative groundedness scoring against structured institutional data

5. Proposed System Architecture



A student query triggers the LLM Response Generator to draft a candidate answer. The Claim Extractor splits this draft into atomic, independently checkable statements. Each claim is routed to three parallel verification agents: a Fact-Check Agent comparing claims against official university records (exam cell data, timetables), a Policy-Compliance Agent checking claims against a deterministic academic policy rule base, and a Grounding Agent checking claims against a retrieval store of the handbook, circulars, and syllabi. The Groundedness Scoring Algorithm aggregates these three signals into a per-claim confidence score, and the Decision Gate passes, flags, or blocks-and-regenerates each claim, with blocked claims routed back to the generator with the verification failure reason attached.

5.1 Groundedness Scoring — Formulation to Develop

Formalize a scoring function $G(c) = w1 \cdot F(c) + w2 \cdot P(c) + w3 \cdot D(c)$, where F, P, and D are normalized [0,1] scores from the fact-check, policy-compliance, and grounding checks respectively, and w1, w2, w3 are weights calibrated against the human-evaluation dataset. Deriving and justifying this weighting — and confirming it transfers across both the NITJ and secondary datasets without re-tuning — is the central technical and generalizability exercise of the project.

6. Dataset & Cross-Institution Validation Design

A single-institution dataset invites the reasonable reviewer objection that results may not generalize. This project addresses that directly by evaluating on two independent sources:

Dataset	Source	Role
Primary: NIT Jamshedpur	Real circulars, academic handbook, exam-cell records, syllabi	Claims labeled true / false / unverifiable by annotators against timestamped institutional records
Secondary: Public university FAQ/QA corpus	Openly available university chatbot or FAQ datasets (non-NITJ)	Cross-institution validation — demonstrates the method generalizes beyond a single university's data

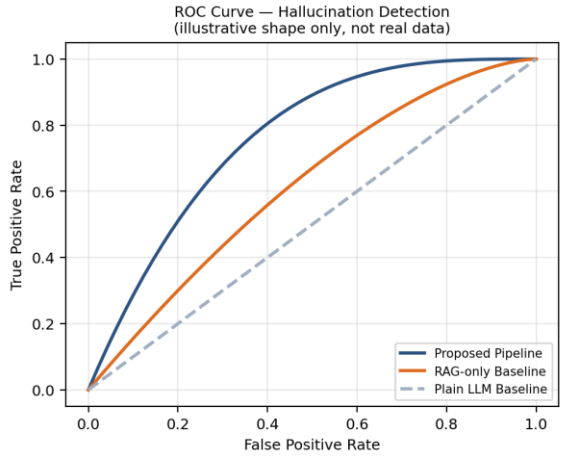
Both datasets follow the same annotation protocol: claims labeled true / false / unverifiable against ground truth by annotators, enabling an apples-to-apples cross-institution comparison rather than two incomparable evaluations.

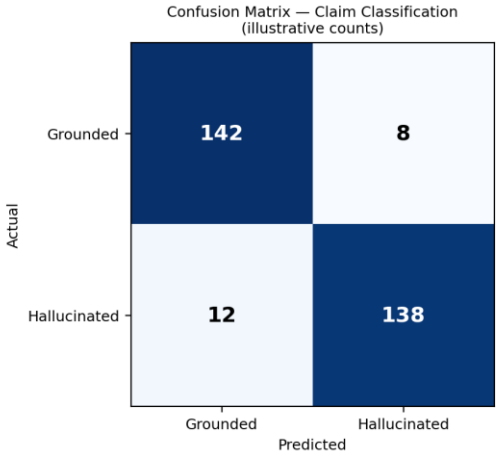
7. Experimental Design

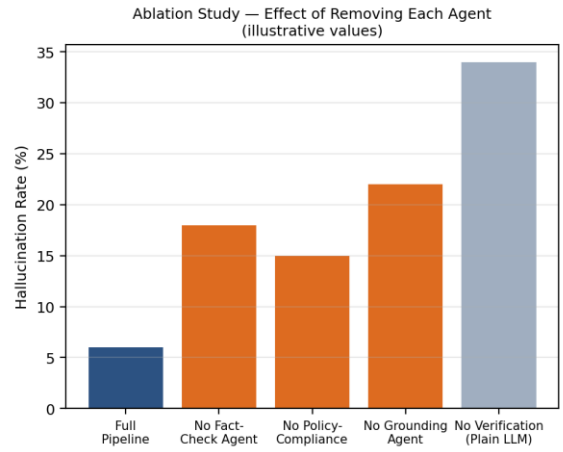
Condition	Description
Baseline A: Plain LLM (no verification)	Direct LLM answer generation with no fact-checking layer
Baseline B: RAG-only	LLM + retrieval grounding, no live-record or rule verification
Baseline C: Self-consistency checking	LLM cross-checks its own claims via repeated sampling (no external verification)
Proposed: Full Verification Pipeline	LLM + claim extraction + fact-check + policy-compliance + retrieval grounding + scoring

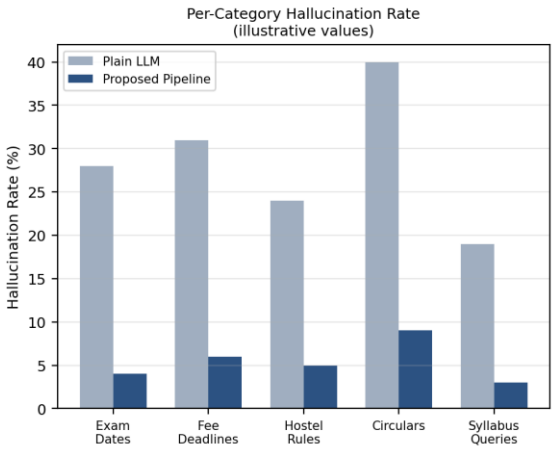
8. Results & Visualization Plan

The figures below illustrate the exact chart types the team should produce once real results are available — shown here with placeholder/illustrative values only, to clarify the expected format, not to represent actual findings.









9. Technology Stack

- LLM Response Generator & Retrieval: LLM API, vector database (Chroma/FAISS) for RAG
- Verification Agents: Python, rule engine for deterministic policy checks
- Scoring & Evaluation: Python, scikit-learn for metrics, pandas, matplotlib/seaborn for result visualizations
- Backend: FastAPI, MongoDB (claim logs, scores, audit trail)
- Frontend: Web dashboard showing claim-level confidence scores and results visualizations

10. Supervision & Meeting Cadence

Weekly review meetings with the supervisor. From Week 9 onward, an additional bi-weekly checkpoint specifically on the scoring algorithm formulation, cross-institution result consistency, and paper draft is recommended, since these require more supervisor involvement than routine implementation work.